Day-6 | AWS Route 53 Explained

1. What is Route 53? Why do we need this service?

Amazon Route 53 is AWS's highly available, scalable, and managed **DNS (Domain Name System) service.** It translates human-friendly domain names such as `www.example.com` into machine-readable IP addresses such as `192.0.2.1`, so browsers and applications can connect to the correct resource. [web:4]

We need Route 53 because users remember domain names, not IP addresses. Route 53 also provides:

- **DNS management** for public and private domains.
- **Domain registration** directly through AWS.
- **Health checks and DNS failover** for highly available applications.
- **Traffic routing policies** such as simple, weighted, latency-based, geolocation, geoproximity, failover, and multivalue answer routing. [web:4]
- **Private DNS** for resolving internal AWS resource names inside a VPC.

Interview one-liner:
"Route 53 is AWS's managed DNS service that routes user traffic to AWS or external endpoints by resolving domain names to IP addresses, with built-in availability, health checks, and intelligent routing."

2. What is DNS? Explain the concept of DNS when using AWS.

DNS (Domain Name System) is the internet's naming system. It converts a domain name like `example.com` into an IP address like `192.0.2.44`, because network communication happens using IP addresses.

DNS in AWS

In AWS, DNS is used to route traffic to resources such as:

- EC2 instances
- Application Load Balancers / Network Load Balancers
- S3 websites
- CloudFront distributions
- API Gateway
- Elastic Beanstalk environments
- RDS databases
- Internal resources inside a VPC

Example:

```
User opens https://app.example.com
        |
        v
DNS lookup through Route 53
        |
        v
Route 53 returns ALB IP address or CNAME/Alias target
        |
        v
User reaches the application behind the ALB
```

AWS also uses internal DNS names for services, for example:

```
my-alb-1234567890.us-east-1.elb.amazonaws.com
mydb.abcdefghij.us-east-1.rds.amazonaws.com
```

These names are useful, but for production you normally create a custom domain such as `app.example.com` and point it to the AWS resource using Route 53.

3. How does the real concept of DNS work with the help of Route 53?

When a user enters `www.example.com`, the following high-level flow occurs:

1. The user's browser needs the IP address for `www.example.com`.
2. The request goes to a DNS resolver, usually provided by the ISP, mobile network, or public DNS provider.
3. If the resolver does not have the answer cached, it queries the root DNS servers.
4. The root server directs the resolver to the `.com` Top-Level Domain (TLD) name servers.
5. The `.com` TLD server returns the Route 53 name servers assigned to `example.com`.
6. Route 53 looks inside the hosted zone for `example.com`.
7. Route 53 finds the record for `www.example.com` and returns the configured value, such as an IP address, CNAME, or Alias target.
8. The browser connects to the returned IP address or endpoint. [web:9]

Example

```
www.example.com  -->  Route 53  -->  ALB DNS/IP  -->  EC2 / ECS / EKS application
```

Route 53 does not usually serve the application itself. It tells the client **where** the application is located.

4. How can we register a domain with Route 53? Why is it easier than the conventional method?

Steps to register a domain

1. Open the **Route 53 console**.
2. Choose **Registered domains**.
3. Choose **Register domain**.
4. Search for the desired domain name, for example `example.com`.
5. Select an available domain and complete the registrant contact details.
6. Review the price and complete the purchase.
7. Route 53 automatically creates a public hosted zone with the same name as the domain.
8. Create DNS records such as `A`, `AAAA`, `CNAME`, `MX`, `TXT`, or Alias records to route traffic.

When you register a domain with Route 53, AWS automatically makes Route 53 the DNS service, creates the hosted zone, assigns four name servers, and updates the domain to use those name servers. [web:8][web:10]

Why Route 53 is easier

Conventional method	Route 53 method
Register domain with one provider	Register domain directly in AWS
Configure DNS with another provider	DNS is automatically configured
Manually update name servers	Route 53 assigns and configures name servers automatically
Separate billing and management	Domain, DNS, AWS resources, and IAM are managed together
More manual integration for AWS resources	Easy Alias records to ALB, CloudFront, S3, API Gateway, etc.

Route 53 automatically creates a hosted zone during registration, reducing manual DNS configuration. [web:2][web:3]

Important: You can also use Route 53 only for DNS management while keeping the domain registered with another registrar. In that case, you create a hosted zone in Route 53 and update the registrar's name-server records to the Route 53 name servers. [web:6]

5. What are Hosted Zones in Route 53? How do they work?

A **hosted zone** is a container for DNS records of a domain and its subdomains. For example, the hosted zone for `example.com` can contain records for:

```
example.com
www.example.com
api.example.com
```

```
dev.example.com
mail.example.com
```

A hosted zone contains information about how to route traffic for a domain and its subdomains. [web:12]

Types of hosted zones

Type	Purpose	Example
Public hosted zone	Resolves DNS from the internet	`www.example.com` accessible publicly
Private hosted zone	Resolves DNS only inside associated VPCs	`db.internal.example.com` for internal applications

Common DNS records

Record	Purpose
`A`	Maps a domain to an IPv4 address
`AAAA`	Maps a domain to an IPv6 address
`CNAME`	Maps a domain/subdomain to another domain name
`Alias`	AWS-specific record that points directly to AWS resources such as ALB, CloudFront, S3, or API Gateway
`MX`	Defines mail servers
`TXT`	Stores text data, commonly used for domain verification, SPF, and DKIM
`NS`	Identifies the authoritative name servers for the hosted zone
`SOA`	Stores administrative information about the DNS zone

For every public hosted zone, Route 53 automatically creates `NS` and `SOA` records. [web:14]

How it works

```
User requests api.example.com
        |
        v
Route 53 public hosted zone for example.com
        |
        v
Finds A/AAAA/CNAME/Alias record for api.example.com
        |
        v
Returns target IP or AWS endpoint
        |
        v
Traffic reaches ALB / CloudFront / EC2 / API Gateway
```

6. High-level flow for Route 53

```
User enters https://www.example.com
                |
                v
Browser asks DNS resolver for IP address
                |
                v
DNS resolver queries root and .com TLD servers
                |
                v
TLD returns Route 53 name servers for example.com
                |
                v
Route 53 hosted zone is queried
                |
                v
Route 53 applies DNS record and routing policy
                |
                v
Returns IP address / AWS endpoint
                |
                v
Browser connects to ALB, CloudFront, S3, EC2, or another endpoint
```

Example production flow

```
www.example.com
      |
      v
Route 53 Alias record
      |
      v
Application Load Balancer
      |
      v
EKS / ECS / EC2 application
      |
      v
Database in private subnet
```

Most Asked Interview Questions and Answers

1. What is Amazon Route 53?

Answer:
Amazon Route 53 is AWS's managed, highly available, and scalable DNS web service. It translates domain names into IP addresses and routes user requests to AWS or external resources. It also supports domain registration, health checks, and traffic-routing policies.

2. Why is it called Route 53?

Answer:
DNS uses TCP and UDP port 53. AWS named the service Route 53 because it routes traffic using DNS on port 53.

3. What is the difference between a CNAME record and an Alias record?

Answer:
A `CNAME` maps one DNS name to another DNS name and cannot normally be used at the zone apex, such as `example.com`. An AWS **Alias record** can point a domain or subdomain directly to supported AWS resources such as an ALB, CloudFront distribution, S3 website, or API Gateway. Alias records are generally preferred for AWS resources because they are free for queries to AWS resources and can work at the root domain.

4. What is a hosted zone?

Answer:
A hosted zone is a container for DNS records related to a domain and its subdomains. A public hosted zone is used for internet-facing DNS, while a private hosted zone is used for DNS resolution within VPCs.

5. What is the difference between a public and private hosted zone?

Answer:
A public hosted zone resolves DNS queries from the internet. A private hosted zone resolves DNS queries only from resources inside the VPCs associated with that hosted zone.

6. What are Route 53 routing policies?

Answer:
Route 53 routing policies determine how DNS responds to a query:

- **Simple routing:** One record or multiple values without special logic.
- **Weighted routing:** Splits traffic by percentage; useful for blue-green or canary deployments.
- **Latency-based routing:** Routes users to the region with the lowest latency.
- **Failover routing:** Routes to a primary endpoint and fails over to a secondary endpoint during failures.
- **Geolocation routing:** Routes based on the user's geographic location.
- **Geoproximity routing:** Routes based on geographic distance, with optional bias.
- **Multivalue answer routing:** Returns multiple healthy records for basic load distribution.

7. What are Route 53 health checks?

Answer:
Health checks monitor endpoint availability from multiple AWS locations. They can be used with failover, weighted, latency-based, and other routing policies to avoid sending traffic to unhealthy endpoints.

8. What happens if Route 53 receives a query for a domain it does not have a record for?

Answer:
Route 53 returns an appropriate DNS response, commonly `NXDOMAIN` if the domain or record does not exist. The application owner must create the required DNS record in the hosted zone.

9. Can Route 53 register a domain and manage DNS separately?

Answer:
Yes. You can register a domain with another registrar and still use Route 53 only for DNS management. You create a hosted zone in Route 53 and update the domain's name servers at the registrar to the Route 53 name servers. [web:6]

10. What is DNS failover?

Answer:
DNS failover automatically routes users from an unhealthy primary endpoint to a healthy secondary endpoint. For example, traffic can fail over from an ALB in `us-east-1` to an ALB in `us-west-2` if the primary region becomes unhealthy.

11. What is the role of TTL in Route 53?

Answer:
TTL (Time To Live) defines how long a DNS response can be cached by resolvers and clients. A lower TTL makes DNS changes propagate faster but increases DNS query volume. A higher TTL improves caching and reduces queries but slows down changes.

12. How does Route 53 improve high availability?

Answer:
Route 53 uses globally distributed DNS servers, health checks, and routing policies such as failover, latency-based, weighted, and geolocation routing. This allows traffic to be redirected away from unhealthy or poorly performing endpoints.